# Business Analytics: risk analysis assignment

*Prof. Reza Skandari, r.skandari@imperial.ac.uk*

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This assignment refers to the TransAtlantic Airlines case study, below. Produce one managerial report (expected around 10 pages) covering both parts, and submit your Jupyter notebook alongside it, built up following the assignment instructions below. Make sure the notebook is clear and easy to review; for simulation outputs, include the key figures and charts in the report too, wherever it refers to them.

## Part I: 70 marks

Assume you have been hired to consult on this case. Give Mike a revised booking policy: your results, your interpretation of them, and a recommendation he can actually act on, stress-tested, not just a number.

## Part II: 30 marks

Mike has a second, more open question for you too, the last one in the brief below. Marks are for the quality and specificity of your idea, your reasoning for why it would actually help, and what it would take to pursue it, not for technical complexity for its own sake. A sharp, well-argued proposal is enough.

# TransAtlantic Airlines: assignment instructions

Read the case first, below.

## The brief

Mike wants a revised booking policy for the Heathrow-JFK route, and a recommendation he can actually act on, not just a pile of numbers.

> "Judy, don't just tell me a booking level. Show me you've actually stress-tested it, and tell me why I should trust it over what we're doing now. One more thing, a competitor just announced a more generous bumping policy. If that changes what bumping a passenger actually costs us in goodwill, I'd rather know now."

Judy's also heard, informally, that gate agents at Heathrow sometimes move a bumped economy passenger into a spare business seat instead of bumping them outright, when there's room. It's never been official policy, just something staff do when it's obviously the right call in the moment.

There's a second question on the table too. Finance has been asking whether TransAtlantic should stop refunding no-shows altogether, something several budget carriers already do. Mike hasn't made up his mind.

> "Judy, if finance gets their way on the refund thing, would that actually help our profit? I don't want a one-line answer, I want to know what you'd check before you'd trust it either way."

One more thing, before you go.

> "Judy, don't let yourself be boxed in by what we're currently measuring. If you had a freer hand, how would you actually make this analysis better, and what would you need, data or otherwise, to get there?"

You also have four months of real boarding records for this route: `heathrow-jfk-boarding-log.csv`. Mike pulled it from ops reporting himself, so treat it the way you'd treat any data handed to you on a real engagement: check it before you trust it. Do an exploratory descriptive analysis before you use any of it.

**One thing worth knowing before you start:** we talked about two types of uncertainty, aleatoric and epistemic. Can you identify both here, and incorporate them into your analysis and report?

# Case Study: TransAtlantic Airlines

### Background information

Airlines were pioneers in the use of risk analysis and Monte Carlo simulation and still make intensive use of these tools, with American Airlines generally considered to be the leader. An important problem that airlines are facing is no-shows: passengers who made a reservation but do not show up for their flights. This results in empty seats and (huge) losses for the airline, considering that whether a passenger actually boards a plane or not, it hardly makes any difference in the operating costs. This has led to the practice of flight overbooking, which means that more reservations are accepted on a flight than the actual capacity of the flight itself. This results in higher flight utilisation rates, at the expense of customers sometimes being "bumped" to the next flight if more passengers show up than can actually board the plane, which calls for some kind of compensation. American Airlines has estimated that the practice of overbooking flights alone adds $200 million per year to its bottom line.

Judy is looking at the figures on the sheet in front of her. Mike, her supervisor who is in charge of yield management at TransAtlantic Airlines, has just called her into his office and told her that the booking policy TransAtlantic has been using for the last two years is urgently in need of revision.

## TransAtlantic Airlines

TransAtlantic operates flights on several routes, mainly connecting London Heathrow to the East Coast of the United States. The sheet contains data on the London Heathrow to New York JFK route, a popular business route that is generating more revenue than any other route in TransAtlantic's portfolio. Every day, 4 flights depart from London Heathrow for New York JFK using a Boeing 747 which can carry up to 500 passengers, 400 in economy class and 100 in business class. The (one-way) ticket price is £450 in economy class and £1,450 in business class. Unlike other airlines, which use many fares with different restrictions (advance purchase, maximum stay, Saturday/Sunday night included, etc.), TransAtlantic charges just one fare per class.

A problem with this route, as well as with other routes, is that often people who have made a reservation do not show up for their flight. TransAtlantic's policy is that both economy and business class passengers who do not show up are refunded.

Since the incremental cost of carrying an extra passenger on a plane is negligible, an empty seat represents a loss equal to the ticket price. In order to make sure that the plane does not fly with too many empty seats, TransAtlantic uses an overbooking policy. However, this overbooking policy sometimes leads to more passengers showing up than can actually get on the plane, resulting in passengers being "bumped" to the next flight. Bumped customers are compensated by being offered a discount voucher, a meal allowance and a hotel room if the next flight leaves the next

day. Bumping also creates a loss of goodwill: some customers might switch to a competing airline for their future flights. Although this cost is difficult to estimate, TransAtlantic believes the cost of bumping a passenger to be £150 for an economy class passenger, and £300 for a business class passenger, including actual costs and goodwill loss. This figure is not a refund: a bumped passenger already paid their fare and still flies, just on a later flight.

## The London Heathrow to New York JFK route

Judy looks at the figures at the bottom of the sheet. The current policy of TransAtlantic Airlines is to accept up to 420 reservations for economy class (i.e. 5% overbooking) and up to 120 for business class (i.e. 20% overbooking). Overbooking in the business class is higher than overbooking in the economy class because the no-show phenomenon is more pronounced in business class.

Judy is not sure why Mike chose exactly this route, perhaps because it is the most important route for TransAtlantic, perhaps because it is one of the routes that are plagued by a high number of no-shows. Whatever the reason, Mike wants her to come up with a revised booking policy for this route, and if things work out well, he would revise the booking policies for the other routes in a similar way.

On the sheet it also says that, until four months ago, TransAtlantic had never kept detailed records about the number of no-shows on the Heathrow-JFK flight. Ops has since started logging boarding numbers flight by flight, but for the years before that, all anyone at TransAtlantic has to go on is experience and judgment. Although the number of no-shows is highly variable from flight to flight, there is a rough consensus that the percentage of no-shows in economy class is somewhere between 3% and 8%, and between 15% and 30% in business class, with 5% and 20% as the most-likely values. Judy notices this is the same range her predecessor used on last year's sheet, she hasn't had a chance yet to check it against the new boarding-log data. Well, Judy thinks, the booking policy seems to make sense then!

Based on his own judgment calls about typical no-show rates, her predecessor's own calculation put the total cost of the current policy, lost revenue on empty seats plus the cost of bumping, at **£6,250** across both classes.